

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-founder

Oct. 2025 - Present

East Lansing, MI

- Developed member portal for hosting projects, classes and events, generating attendance QR codes stored in S3 with CloudFront CDN for low-latency delivery, and fully automating GitHub Teams assignment and Slack invitations for 120+ members, reducing manual setup time from 15 minutes to <30 seconds per member.
- Built the platform with open source best practices, implementing zero-downtime database migrations and maintaining strict 85% code coverage and linting standards via CI/CD (GitHub Actions) and testing suites.
- Optimized infrastructure for horizontal scaling, implementing connection pooling (PgBouncer) and stateless API design to support 100+ concurrent users with consistent sub-200ms response times.

MindMosaic | *React, Vite, FastAPI, PostgreSQL, Neo4j*

Product Manager

Aug. 2025 – Nov. 2025

East Lansing, MI

- Implemented OAuth 2.0 authentication with Google and GitHub providers, enforcing 15-minute access token and 7-day refresh token rotation with httpOnly cookie session management.
- Architected attribute-based access control (ABAC) with PostgreSQL-backed permission store supporting 3-tier role hierarchy (admin/member/viewer), implementing middleware-based authorization checks.
- Optimized PostgreSQL schema with compound B-tree indexes on (user_id, created_at), reducing note search latency from 850ms to 440ms and cutting database CPU utilization by 30% under 50 concurrent queries.

PROJECTS

Mira | *Next.js, Electron, FastAPI, Redis, PostgreSQL*

End-to-end voice intelligence system, aggregating real-time audio streams to execute context-aware automation

May 2025 – Nov. 2025

- Architected containerized microservice deployment with Docker Compose, orchestrating FastAPI backend, PostgreSQL database, and Redis cache across isolated containers with shared Docker network and persistent volume mounts, integrating Electron desktop client with bi-directional IPC for real-time audio streaming.
- Built production hybrid search pipeline merging BM25 keyword matching with Sentence-BERT embeddings (384-dim), implementing 5-minute conversation boundary detection across multi-stream audio ingestion to optimize context retrieval accuracy for RAG system.
- Designed RESTful API with FastAPI using Pydantic schema validation and OpenAPI 3.0 documentation, implementing lazy-loaded dependency injection for heavy ML models and centralized logging with custom exception middleware for consistent error handling.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy*

Won Interactive Media Track with real-time computer vision game utilizing pose-estimation and gesture-detection

Feb. 2025

- Engineered a low-latency gesture pipeline using MediaPipe Pose to map 33-point skeletal landmarks to game actions, achieving 60 FPS with sub-20ms input lag by optimizing the inference loop.
- Developed a custom vector-based physics engine handling AABB collision detection to map 3D real-world movements into 2D game state in real-time.

TECHNICAL SKILLS

Cloud & Infrastructure AWS (EC2, RDS, S3, CloudWatch), Docker, GitHub Actions, PostgreSQL

Backend Frameworks FastAPI, SQLAlchemy, Redis, Alembic, Node.js

Frontend Development React, Next.js, TypeScript, Tailwind CSS, Electron, Vite

Languages Python, C++, Java, JavaScript/TypeScript, SQL, Bash

